Understandings of dealing with my dads gift to me, which was sadly a fake:

So i understand a few things happened with this device. It had firmware spoofing on windows the storage etc. It broke in 2 when we went to set the new file system type (Another red flag) And it wouldnt load the repair of damaged partitions as they werent there to begin with

Poh: Yes — your understanding is pretty solid now. What you observed lines up very closely with how counterfeit/fake-capacity storage devices behave.

Here's the technical breakdown of what likely happened:

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1. Firmware spoofing / fake capacity reporting

The controller firmware inside the device falsely reported a huge capacity to the OS.

So Windows initially believed things like:

- 15 TB
- 64 GB
- etc.

because operating systems generally trust what the storage controller advertises.

But the actual physical flash memory inside was far smaller.

F3 proved that directly:

```
Usable size: 11.01 GBAnnounced size: 62.50 GB
```

That is classic fake-capacity firmware spoofing.

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2. The disk "splitting in two"

This was another major red flag.

What likely happened:

- the fake controller could not correctly manage partition structures at the advertised size,
- GPT backup structures near the end of the disk failed,

- Windows/Linux started seeing inconsistent geometry,
- and partition metadata became corrupted/confused.

That’s why you saw:

- split unallocated sections,
- inconsistent capacities,
- unstable partition layouts.

A genuine healthy SSD generally does NOT randomly fragment itself during basic formatting operations.

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3. Partition repair failures

Exactly.

Tools could not properly repair the filesystem/partition structures because:

- the underlying storage geometry itself was false or unstable,
- sectors beyond the real flash capacity didn’t genuinely exist,
- and the controller was lying about accessible address space.

That’s why Linux kept throwing:

```
Logical block address out of range
```

The OS tried accessing sectors the firmware claimed existed, but the hardware could not actually service them.

So repair tools were essentially trying to rebuild structures on nonexistent or invalid storage space.

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4. Why Windows initially “worked”

This is important.

Fake drives often:

- appear normal at first,
- allow small file copies,

- mount correctly,
- even format successfully.

Because:

- the first few GB of real flash memory DO exist.

The corruption usually starts:

- once writes exceed the real capacity,
- or when partition tools touch higher LBAs.

That’s why many people don’t discover the scam immediately.

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5. Why Linux exposed it faster

Linux tends to:

- expose kernel-level storage errors more directly,
- validate block behavior more strictly,
- and provide better low-level diagnostic tools.

Especially:

- `dmesg`
- `smartctl`
- `f3probe`

Those tools revealed the contradictions very clearly.

Your core insight is correct

The important realization is:

> The problem was not just a damaged filesystem.
> The underlying storage reporting itself was fraudulent or unstable.

That’s why:

- formatting behaved strangely,
- partitions became inconsistent,

- repair attempts failed,
- and geometry changed unpredictably.